ODS: Test-Time Adaptation in the Presence of Open-World Data Shift

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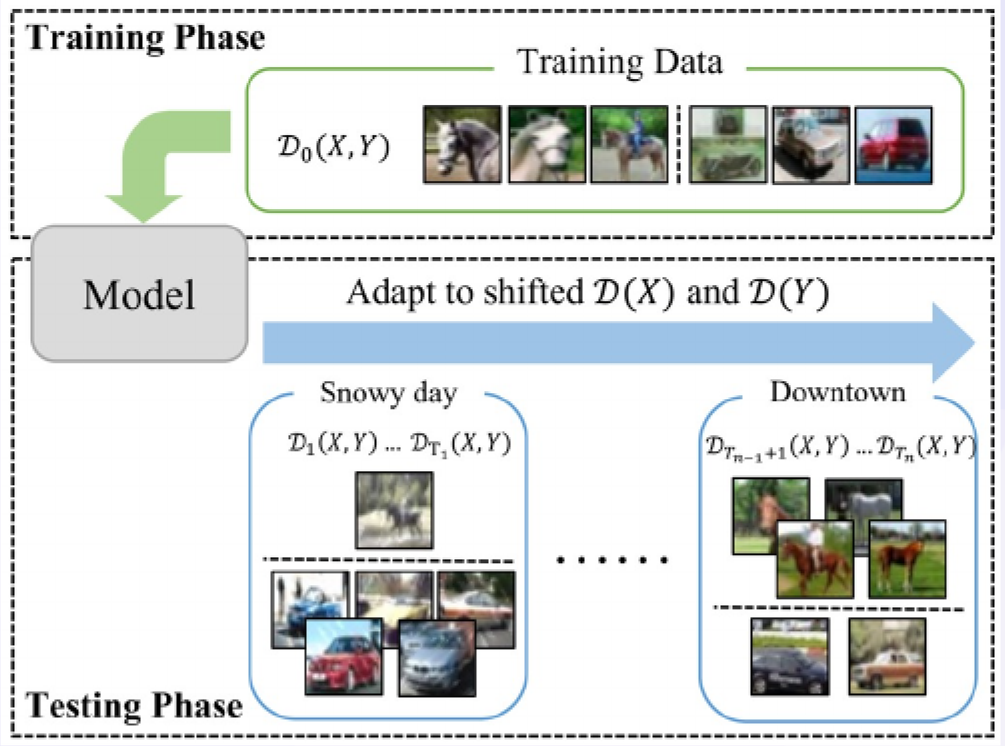
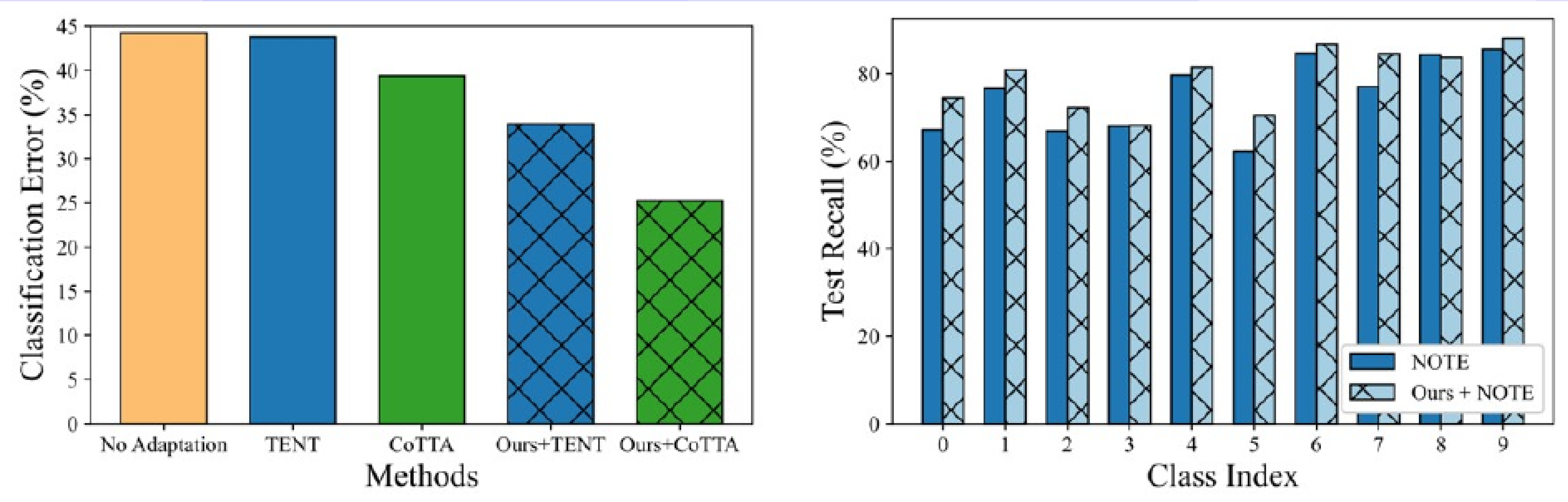
01 ODS: Test-Time Adaptation under Open-World Data Shift — Authors



- We present a framework for robust test-time adaptation.
- We study open-world shifts with changing inputs and labels.

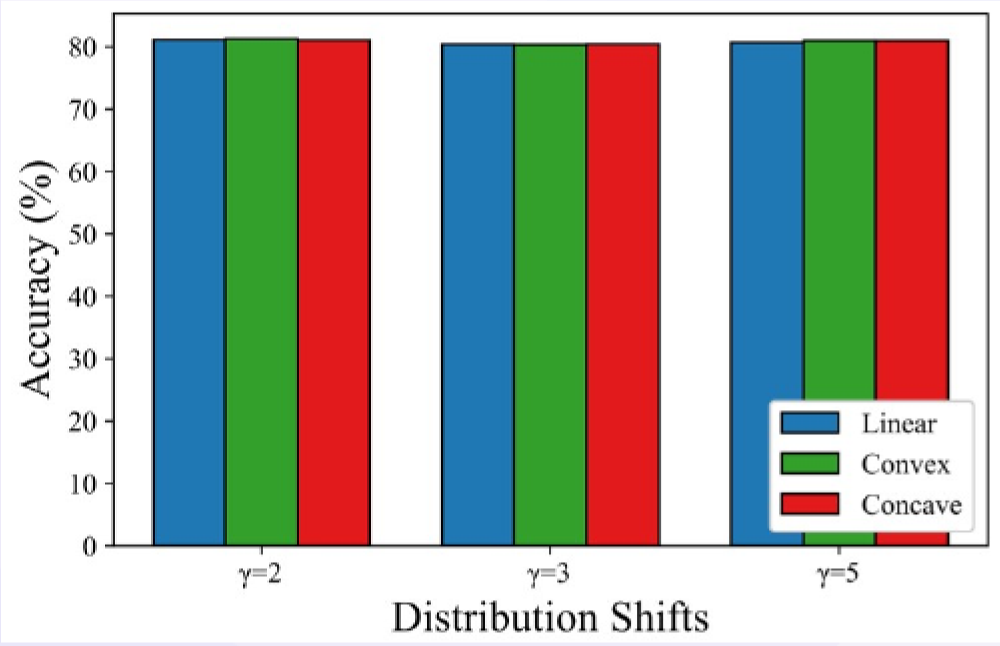
02 Introduction: Why ODS for Open-World TTA



(a) Classification Error (%) (b) Test Recall(%)

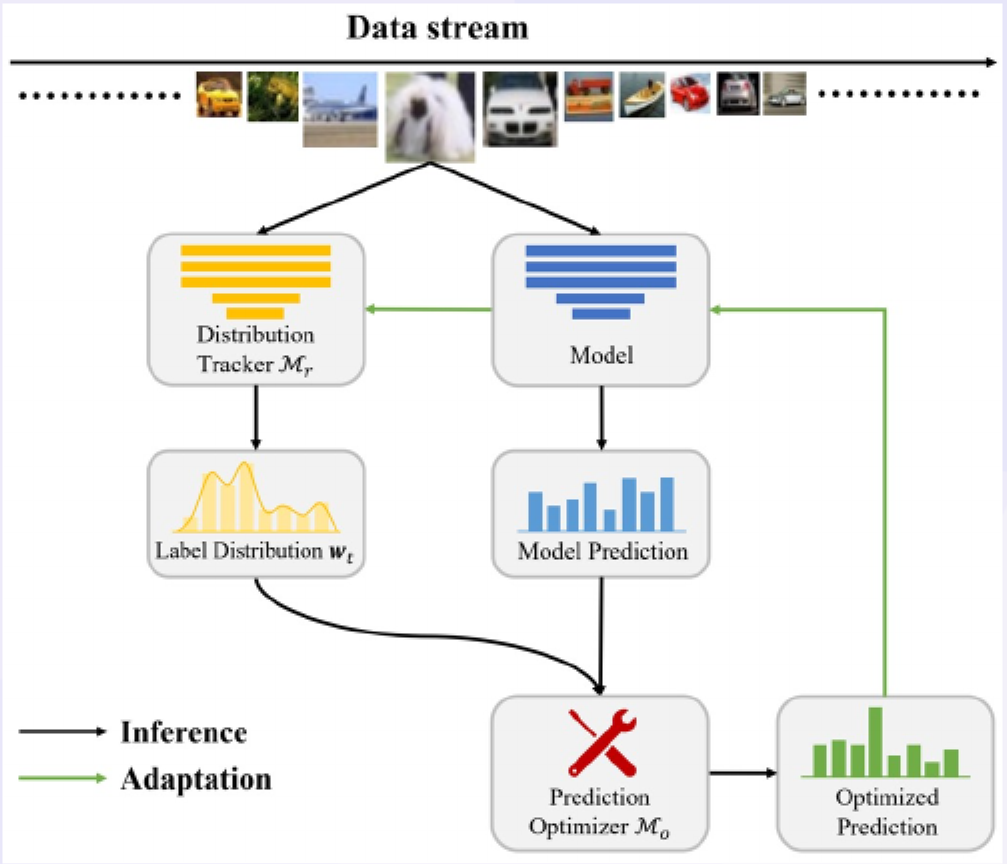
- We study AODS with shifts in inputs and labels over time.
- We show existing TTA overlooks label shifts and degrades accuracy (e.g., BN STATS (Ioffe & Szegedy, 2015), TENT (Wang et al., 2021), EATA (Niu et al., 2022), COTTA (Wang et al., 2022), LAME (Boudiaf et al., 2022), NOTE (Gao et al., 2023)).
- We track the label distribution and optimize predictions.
- We plug into many backbones and obtain consistent gains.

03 Problem Setup and Error-Bound Insights under GLS



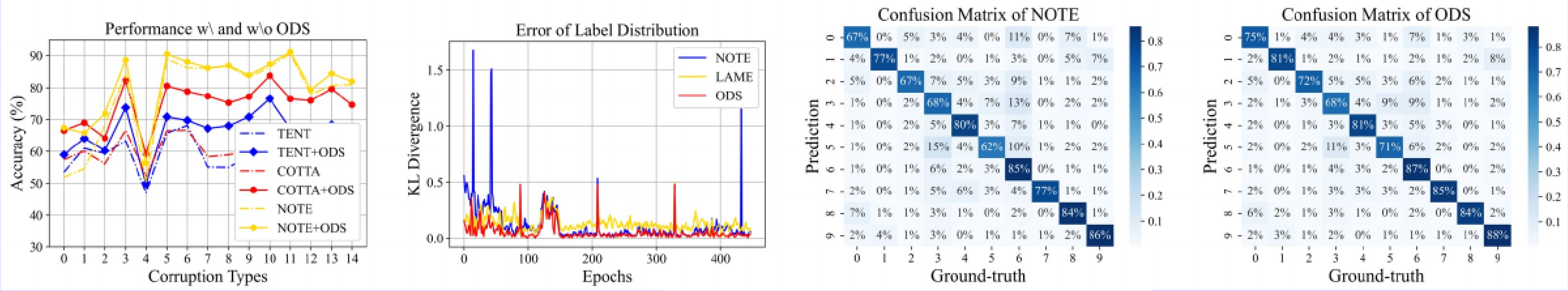
- We consider multi-class streaming adaptation without test labels.
- We adopt generalized label shift: features given class are stable (prior GLS; e.g., Lipton et al., 2018; Azizzadenesheli et al., 2019).
- We show error decreases with better label tracking and features.
- We bound error by label-mismatch and feature misalignment; we therefore track w_t and adapt features.

04 Method: ODS Overview — M_T (Tracking) + M_O (Prediction Optimization)



- We estimate w_t from unlabeled data with M_T.
- We refine predictions using w_t via M_O.
- We design a backbone-agnostic procedure with stable updates.

05 Experimental Setup and Research Questions (with baseline citations)



- We ask RQ1–RQ3 on effectiveness, generality, and estimation quality.
- We use CIFAR10/100-C (severity 5) and induce label skew via tweak-one with γ∈{2,5,10}.
- We compare to BN STATS (Ioffe & Szegedy, 2015), TENT (Wang et al., 2021), EATA (Niu et al., 2022), COTTA (Wang et al., 2022), LAME (Boudiaf et al., 2022), NOTE (Gao et al., 2023).
- We cite all baselines in-line on result slides.

06 Evaluation Metrics for Main Results

- We report Top-1 accuracy = fraction of samples whose argmax prediction is correct.
- We report Per-class accuracy = mean of per-class accuracies to assess class balance.
- We compute KL(w_t || D_t(Y)) to assess label-estimation error.
- We measure Runtime = per-batch wall-clock for adaptation.

07 Main Results and RQ**2**/RQ**3** Highlights (with baseline citations)

Methods	Noise			Blur				Weather				Digital				Avg.
	Gauss.	Shot	Impul.	Defoc.	Glass	Motion	Zoom	Snow	Frost	Fog	Brit.	Contr.	Elastic	Pixel	JPEG	
Source	14.70	18.52	15.61	56.92	31.99	68.01	63.25	82.19	72.44	76.31	**92.41**	23.38	72.33	68.72	79.72	55.77
Bn Stats	50.60	51.16	45.31	71.73	47.99	69.35	68.59	60.16	60.39	64.27	69.60	67.56	59.21	66.12	58.17	60.68
Tent	53.53	60.97	59.34	63.33	47.12	65.81	68.11	55.08	55.00	58.68	63.40	49.59	46.95	50.45	45.38	56.18
Eata	48.94	48.21	42.05	65.44	43.42	59.81	57.27	55.09	52.98	56.00	59.54	61.47	51.32	55.75	50.88	53.88
Lame	57.99	60.15	53.07	78.83	53.04	76.67	74.90	67.81	67.30	71.94	77.05	74.84	68.53	73.44	66.90	68.16
CoTTA	57.43	60.06	56.03	66.66	52.25	66.54	66.65	58.32	58.92	60.09	64.69	55.05	59.37	64.74	61.92	60.58
Note	51.90	54.57	68.38	84.29	50.53	88.97	86.21	86.15	86.68	83.27	86.48	90.64	77.84	80.77	81.02	77.18
Ods	**67.45**	**65.78**	**71.88**	**88.66**	**56.32**	**90.48**	**88.09**	**86.16**	**86.93**	**83.96**	87.37	**91.16**	**79.35**	**84.43**	**82.02**	**80.67**

- We achieve best averages across severities and corruptions (Table 1).
- We improve TENT (Wang et al., 2021), COTTA (Wang et al., 2022), and NOTE (Gao et al., 2023).
- We find distribution optimization outperforms statistics optimization.

08 M_T Deep Dive: Distribution Tracking and Ablations

Modules $\mathcal{M}_T$	Modules $\mathcal{M}_O$	TENT	CoTTA	NOTE
		56.18 ± 4.13	60.58 ± 0.15	77.18 ± 0.38
✓		58.95 ± 2.36	60.65 ± 0.31	77.20 ± 0.57
✓	✓	**66.03 ± 1.89**	**74.72 ± 0.64**	**80.67 ± 0.29**

- We estimate labels by combining entropy and feature consistency.
- We smooth w_t over time with a FIFO queue.
- We show ablations vs TENT (Wang et al., 2021), COTTA (Wang et al., 2022), NOTE (Gao et al., 2023); full ODS performs best.

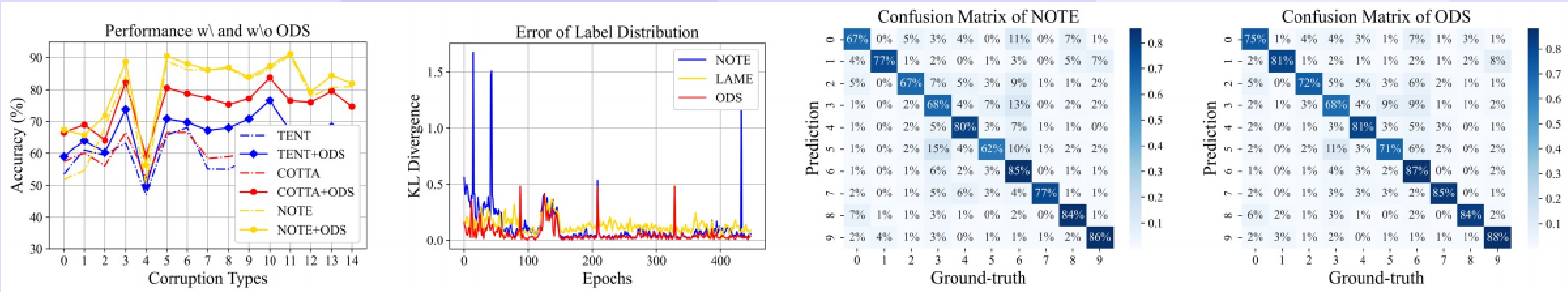
09 Limitations and Open Questions

- We are not state of the art without label shift in some cases.
- We may be sensitive to noisy w_t and many classes at scale.
- We have not extended to detection and multimodal streams.
- We have not audited fairness under shift and will study disparate impact.
- We risk catastrophic forgetting under long streams and will explore regularizers/checkpointing.

10 Related Work (Concise Positioning with Baseline Citations)

- We relate to fully test-time TTA: BN STATS (Ioffe & Szegedy, 2015), TENT (Wang et al., 2021), EATA (Niu et al., 2022); continual TTA: COTTA (Wang et al., 2022), LAME (Boudiaf et al., 2022), NOTE (Gao et al., 2023).

11 Conclusion and Key Takeaways



- We track label distribution and optimize predictions effectively.
- We plug into many TTA methods and obtain consistent gains under open-world shifts.
- We scale across methods and datasets.

Thank You

Questions & Discussion